INCH-POUND

MIL-C-3098/51F
10 April 1995
SUPERSEDING
MIL-C-3098/51E
17 July 1984

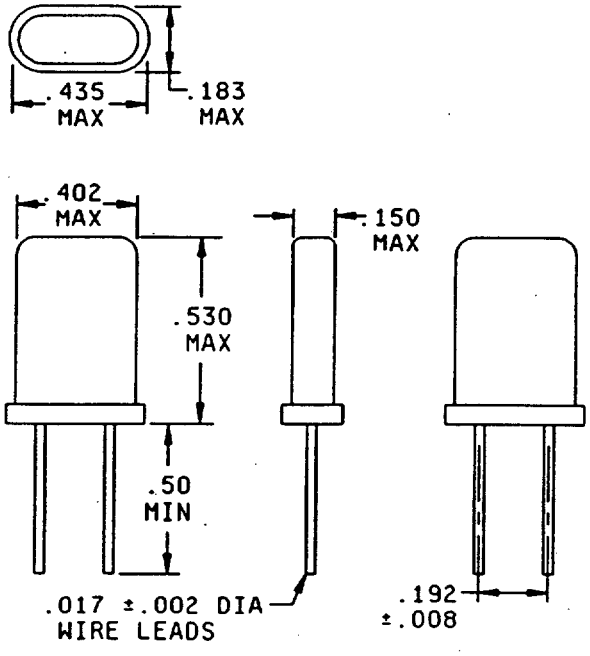
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-74/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 45 MHz to 125 MHz; fifth overtone; controlled; series.



Inches	mm
.002	0.05
.008	0.20
.017	0.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-C-3098.
4. If the HC-43 holder is used the crystal unit shall not be trimmed after cold-welding.

FIGURE 1. Crystal unit - CR-76/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-26/U (MIL-H-10056/13) or HC-43/U (MIL-H-10056/9) or HC-49/U (MIL-H-10056/36).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 45 MHz to 125 MHz, inclusive.

Capacitance, shunt: 4.5 pF ±0.5.

Frequency tolerance:

Operating temperature range: ±10 ppm.

Frequency stability: ±2.5 ppm.

Equivalent resistance: 20 ohms, minimum; 50 ohms, maximum.

Mode of oscillation: Fifth mechanical overtone.

Temperature ranges:

Operating: +80°C to +90°C, inclusive.

Operable: -55°C to +80°C, inclusive.

Reference temperature: +85°C ±1°C.

Drive level, rated: 1.0 ±0.2 mW.

Unwanted mode: Twice main mode.

Shock (specified pulse):

Frequency change permitted: ±3 ppm.

Equivalent resistance change permitted: ±15 percent.

Vibration high frequency: MIL-STD-202, method 204, test condition A.

Frequency change permitted: ±3 ppm.

Equivalent resistance change permitted: ±15 percent.

Thermal shock:

Frequency change permitted: ±3 ppm.

Equivalent resistance change permitted: ±15 percent.

Aging:

Frequency change permitted: ±3 ppm.

NOTE: NATO guidance type NXT50.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 17, 19, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0680-01)

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